Python for Data Science

-Final Project-

SpaceX Falcon 9 Launch Success Analysis

August 12, 2026

Executive Summary

This project analyzes SpaceX Falcon 9 launch data to identify key factors influencing launch success rates. Data was gathered from Wikipedia using **Beautiful Soup** and **Pandas**, and analyzed using **SQL**.

Method

Various machine learning models—Logistic Regression, SVM, Decision Tree, and KNN—were implemented and compared to predict landing outcomes.

Findings

Launch success rates increased significantly from **2013 onward**, reaching around **80%** in recent years.

Different launch sites exhibited varying success rates, with some sites achieving higher payload capacities than others.

Certain orbit types, such as ES-L1 and GEO, had **100% success rates**, while others, like SO, showed a **0% success rate**.

Although high-payload launches were less frequent, they had a higher success proportion, indicating their reliability.

The Decision Tree model achieved the highest test accuracy of **94.4%** in predicting landing outcomes.

These insights contribute to a deeper understanding of the reliability and efficiency of reusable launches, guiding future improvements in SpaceX operations and informing strategic decisions for upcoming missions.

Executive Summary

Data Collection and transform to DataFrame

- **Wikipedia:** Web scraping was performed using **Requests and Beautiful Soup** to extract launch records from HTML tables.
- **SpaceX API:** requests.get() was used to retrieve JSON data, which was converted into a DataFrame using **Pandas json_normalize()**.
- Information from both sources was combined into a dataset for further analysis.

Perform Data Wrangling

EDA (Explanatory Data Analysis) and SQL

- **SQL** was used to query, aggregate, and analyze the launch dataset.
- **Matplotlib and Seaborn** were used to visualize relationships between launch success and key variables.

Interactive visual analysis

- **Folium** was used to map launch sites and examine geographical patterns.
- **Plotly Dash** was used to create interactive visualizations for **launch site and payload analysis**.

Predictive analysis

- Standardized numerical features using **StandardScaler** and split the data into **80% training and 20% test sets**.
- Trained and compared **Logistic Regression, SVM, Decision Tree, and KNN** models.
- Used **GridSearchCV with 10-fold cross-validation** to optimize hyperparameters.

Introduction

- SpaceX has conducted numerous Falcon 9 launches, making the successful landing and reusing of boosters crucial for enhancing reusability and reducing launch costs. This innovative approach is essential for advancing space exploration and commercial spaceflight.

- However, the outcomes of launches and landings can vary significantly based on factors such as launch site, payload mass, and orbit type. Understanding these factors is vital for improving the reliability and efficiency of reusable launches, ultimately contributing to the sustainability of space missions.

- In this project, I analyzed historical Falcon 9 launch data to identify patterns related to launch and landing outcomes. Additionally, I built and compared various machine learning models to predict the success of booster landings, aiming to provide valuable insights for future launches.

SpaceX API (1)

Two data collection methods were used: web scraping was used to collect launch records from Wikipedia, while the SpaceX API was used to obtain detailed information linked to launch IDs.

SpaceX API
- requests.get() — HTTP GET Request

JSON Response
- pd.json_normalize() — JSON → DataFrame

Select relevant columns
- rocket / payloads / launchpad / cores / flight_number / date_utc

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

SpaceX API (2)

Extract information from ID-based fields

- rocket → Booster Version
- payloads → Payload Mass / Orbit
- launchpad → Launch Site / Latitude / Longitude
- cores → Landing Outcome / Flights / GridFins / Reused / Legs / Landing Pad

Data Filtering

- Remove launches other than Falacon9
- Then Create final Pandas DataFrame

Handle missing values

- Payload Mass: replace missing values with mean

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

SpaceX API (3)

The following is the processed data:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
0	1	2006-03-24	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
1	2	2007-03-21	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
2	3	2008-09-28	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
3	4	2009-07-13	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
4	5	2010-06-04	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection (Web Scraping)

HTTP GET Request

- requests.get() from Wikipedia

It allows us to access and handle data from websites.

HTML Response

- BeautifulSoup
- This allows for easy navigation and searching within the HTML structure

Find Target Tables

- find_all("table")
- Identified the target table and get returns

Extract Table Headers & Records

- find_all("th") / find_all("tr")
- Retrieve table headers and records

Pandas DataFrame

- pd.DataFrame()
  - Date and Time Separation: Initially, date and time were combined in the same record. They were separated into two distinct columns—date and time—to facilitate analysis.

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20(1).ipynb

Data Collection (Web Scraping)

The following is the result of web scraping:

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	F9 v1.07B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0005.1	No	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success	F9 v1.07B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success	F9 v1.07B0007.1	No	1 March 2013	15:10

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20(1).ipynb

Data Wrangling

To ensure data integrity

- Null Value Check: The dataset was examined for null values to quantify their presence and impact on analysis.
- Data Type Verification: The data types were analyzed to distinguish between numerical and categorical data.

Data Transformation

- Launch Counts by Site and Orbit: An exploratory analysis was conducted to determine the number of launches and outcomes categorized by launch site and orbit type.
- Outcome Simplification: A new column named "class" was created to represent outcomes:
    - Class = 1: Successful launch
    - Class = 0: Unsuccessful launch

This transformation allowed for easier analysis and interpretation of results. The analysis revealed that the success rate across all launches was 66.7%.

https://github.com/HaruNNH/Final-Projects/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

Data Wrangling Methodology

The following is the result of data wrangling:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

A new column, "class," has been added to the far right, indicating whether the launch was successful or not.

Data Cleaning

Before performing the SQL analysis, records with missing values in the Date column were removed from the dataset. These records were excluded because they did not represent valid launch records. The cleaned data was then stored in an SQLite table, SPACEXTABLE, for further analysis.

https://github.com/HaruNNH/Final-rojects/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb
https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

EDA with Data Visualization (1)

Method

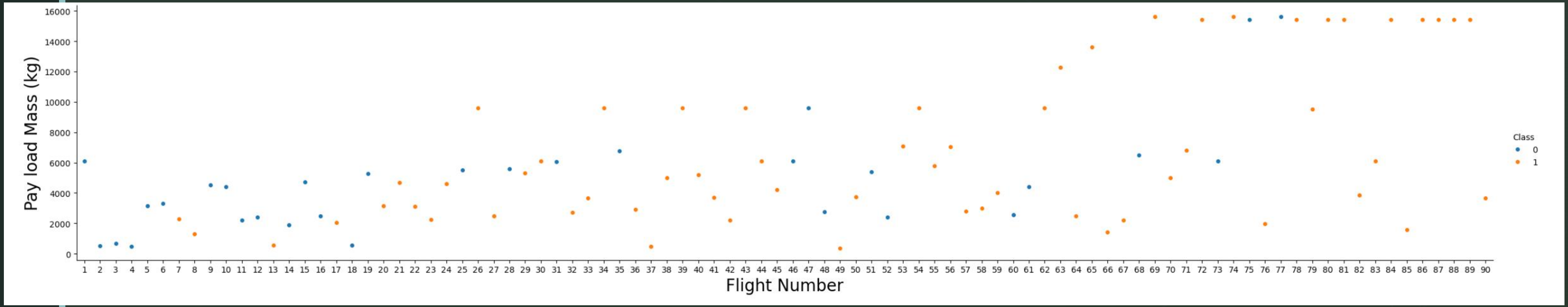
I mainly used **matplotlib** and **Seaborn** to analyze the collected data and visualize the results in graphs.

Purpose:

The scatter plot shows the relationship between payload mass and flight number, with different colors indicating whether the launch was successful or unsuccessful. Examining changes in payload mass and launch success over time.

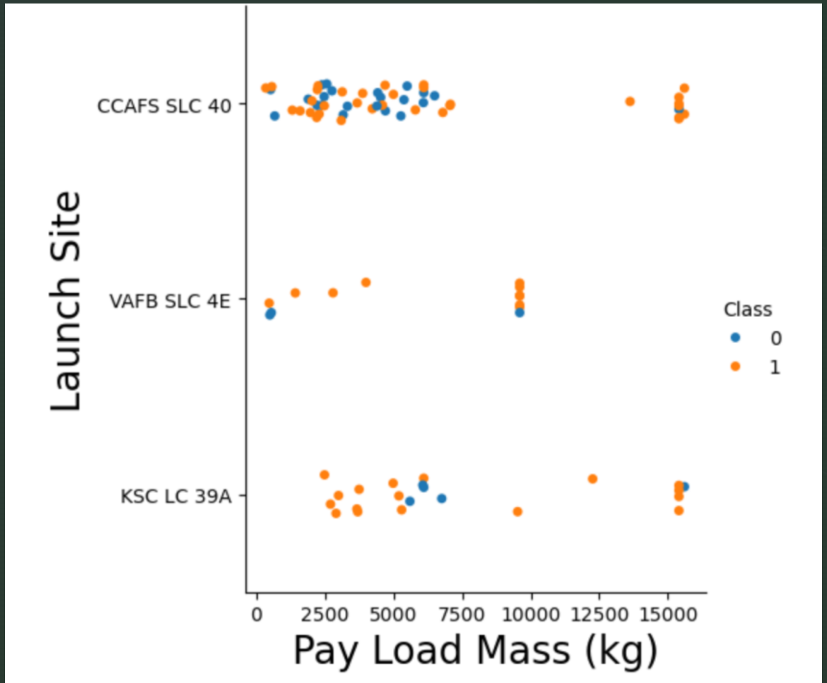
Findings:

As the flight number increases, the payload mass generally becomes heavier. Payloads exceeding 1,600 kg first appear around Flight 70. At the same time, the launch success rate shows an increasing trend over time.



https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

EDA with Data Visualization (2)



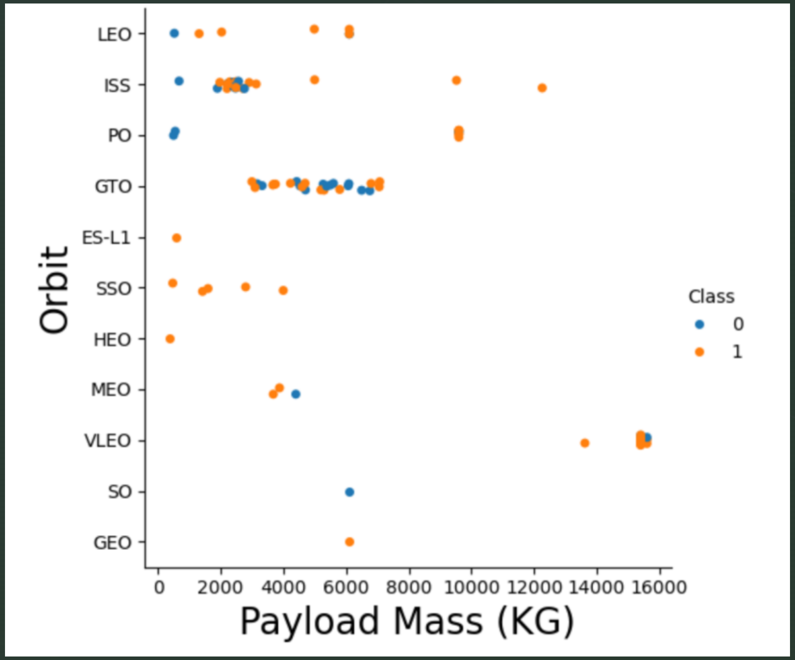
Purpose: The scatter plot illustrates the relationship between payload mass, launch site, and launch outcome (success or failure). This visualization aims to reveal how payload weight and launch site influence launch success rates.

Findings

- Smaller payloads, particularly those weighing less than 7,500 kg, were more common. However, heavier payloads appear to have a higher success rate.
- At **VAFB SLC 4E**, there were no launches with payloads exceeding 10,000 kg.
- **CCAFS SLC 40** had the highest number of launches, but it also had the highest number of failed launches among the launch sites.
- At **KSC LC 39**, launches with payloads below 2,500 kg were relatively rare.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

EDA with Data Visualization (3)



Purpose

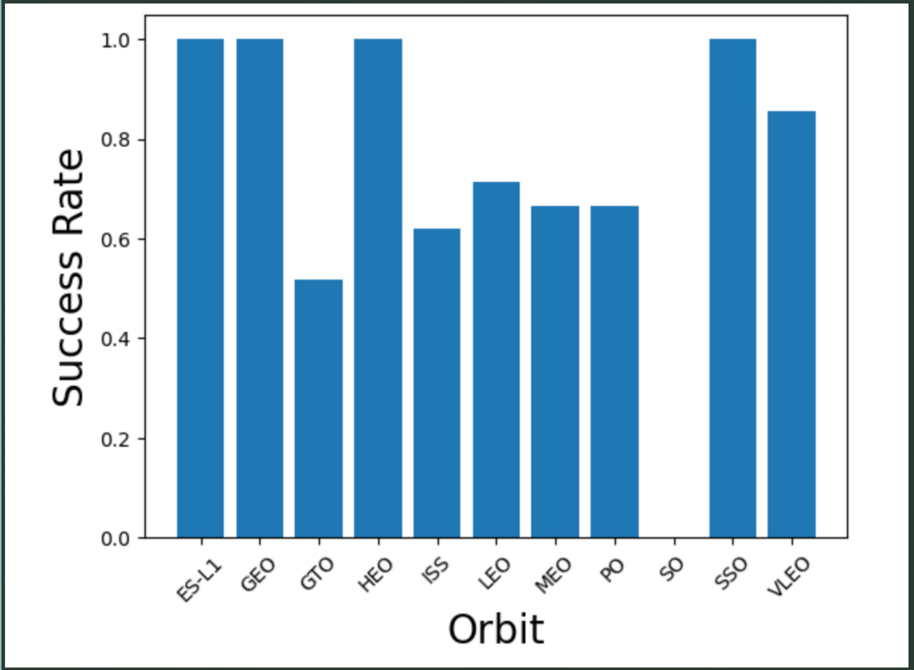
The following scatter plot illustrates the relationship between **payload mass, orbit, and launch outcome (success or failure)**. It aims to show how payload mass and orbit selection affect launch success.

Findings

- **GEO, HEO, and ES-L1** each have only one launch attempt, making clear patterns difficult to identify.
- **GTO** has many attempts and a relatively high failure rate, with payloads mainly between **4,000 and 8,000 kg**.
- **LEO, PO, and ISS** show failures mainly with payloads below **4,000 kg**, while heavier payloads have no observed failures. **SSO** has several attempts, all successful.
- **VLEO** sometimes carries the heaviest payloads, reaching about **16,000 kg**, with successful launches more common than failures.
- Overall, **heavier payloads tend to have higher success rates, while orbit selection also affects launch outcomes.**

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

EDA with Data Visualization (4)



Purpose

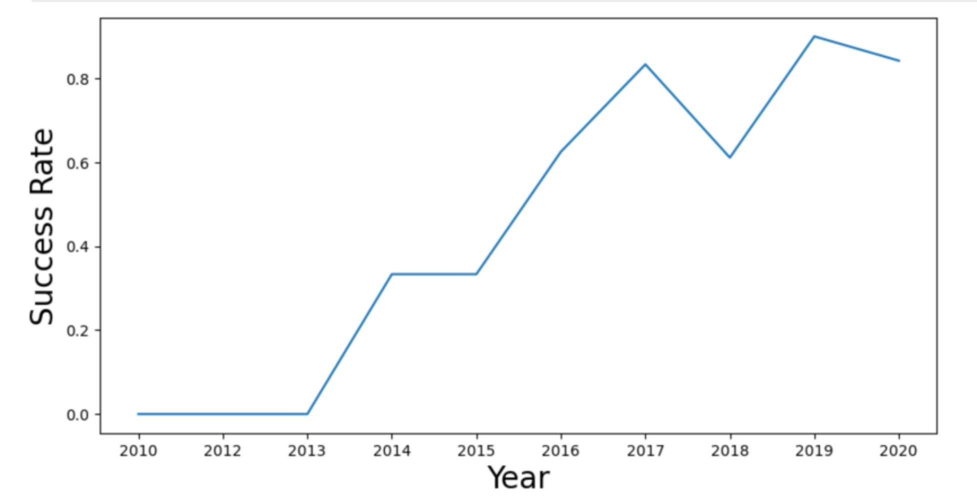
The following bar graph illustrates the relationship between orbit type and success rate. This visualization aims to identify the most appropriate orbit for launch based on the success rates recorded for each orbit, which can inform future launch strategies.

Findings

- The orbits with the highest success rates—ES-L1, GEO, HEO, and SSO—each recorded a 100% success rate, indicating their reliability for successful launches.
- Following these, VLEO achieved an 80% success rate.
- LEO has a success rate of approximately 70%, while MEO and PO have similar rates of slightly over 60%. ISS recorded a 60% success rate, and GTO had a success rate of around 50%.
- Notably, SO had the lowest success rate at 0%, highlighting potential challenges associated with this orbit.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

EDA with Data Visualization (5)



Purpose

The graph shows changes in launch success rates over time, highlighting the improvement in SpaceX's launch reliability. This analysis helps identify trends in performance and factors related to launch success.

Findings

The success rate generally increased from 2013 to 2019, with a significant rise of about 30 percentage points from 2013 to 2014. The rate remained stable in 2015, then increased to around 60% in 2016. Notably, it first reached **80% in 2017**, but experienced a drop to 60% in 2018 before reaching its highest success rate of **over 80% in 2019**.

This fluctuation highlights the challenges faced in 2018, which may warrant further investigation into factors that affected launch outcomes that year. Overall, the trend indicates a positive trajectory in SpaceX's launch success rates over time.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

SQL Analysis Results

Purpose: Display the names of the unique launch sites in the space mission

SQL example:
pd.read_sql("SELECT DISTINCT Launch_Site FROM SPACEXTABLE", con)

Result:
There are four different sites, CCAFS LC-40, VAFB SLC-45, KSC LC-39A, and CCAFS SLC-40.

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Purpose: List the total number of successful and failure mission outcomes

SQL example:
pd.read_sql("""
SELECT "Mission_Outcome", COUNT(*) AS count
FROM SPACEXTABLE GROUP BY "Mission_Outcome"""", con)

Result:
Almost all missions were successful.

	Mission_Outcome	count
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

SQL Analysis Results

Purpose: Display the total payload mass carried by boosters launched by NASA (CRS)

SQL example:
pd.read_sql("SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'", con)

	total_payload_mass
0	45596

Purpose: Display average payload mass carried by booster version F9 v1.1

SQL example:
pd.read_sql("SELECT AVG(PAYLOAD_MASS__KG_) AS average_payload_mass FROM SPACEXTABLE WH

	average_payload_mass
0	2928.4

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

SQL Analysis Results

Purpose: List the date when the first succesful landing outcome in ground pad was acheived.

SQL example:
pd.read_sql("SELECT MIN(Date) AS first_successful_ground_landing FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'", con)

Result:
The first successful ground landing was on December 22, 2015.

	first_successful_ground_landing
0	2015-12-22

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

SQL Analysis Results

Purpose: Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

SQL example:

```
pd.read_sql("""
SELECT "Landing_Outcome", COUNT(*) AS count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY count DESC""", con)
```

Result:

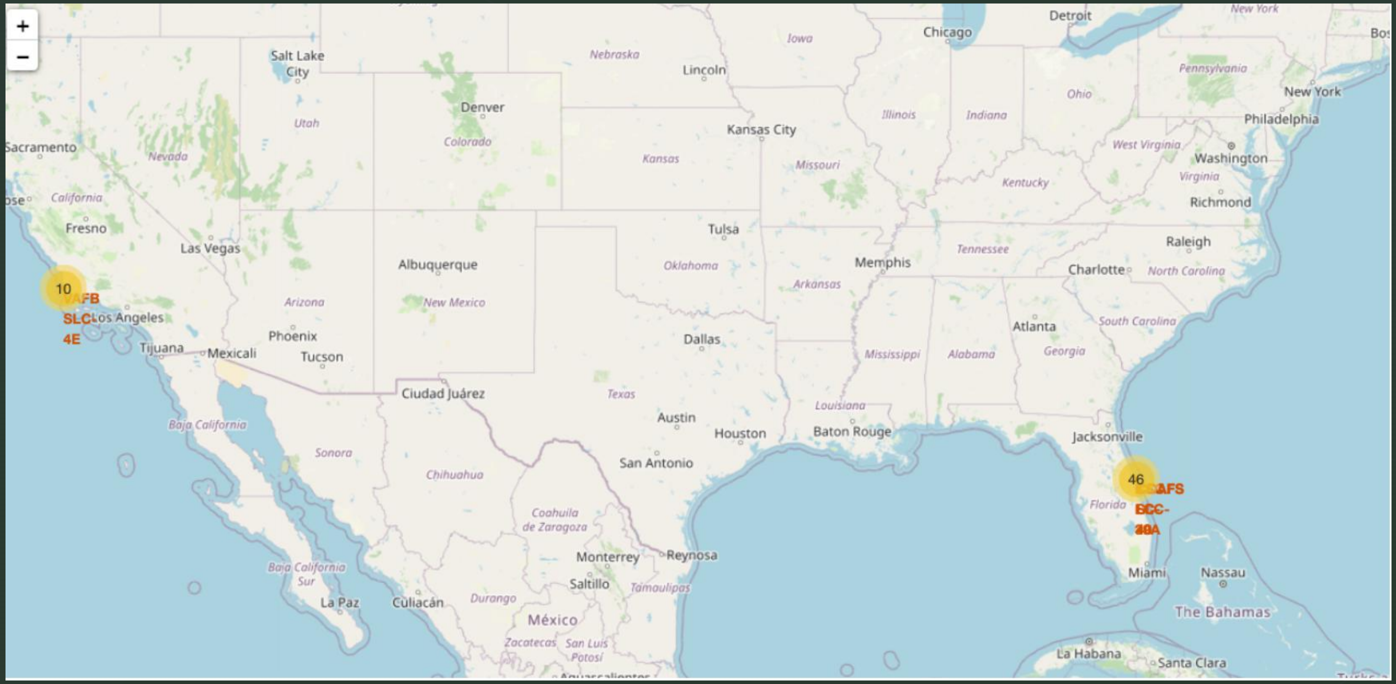
No attempt was the most frequent outcome, with a count of 10, followed by Success (drone ship) and Failure (drone ship), with 5 each.

	Landing_Outcome	count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Folium Map Results

The following map was created using Folium. It contains 56 launch locations, with each record showing the launch site, geographic coordinates, and launch success class. Most launches took place at sites on the East Coast, with 46 launches.
VAFB SLC-4E is located on the West Coast, while the others—CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A—are located on the East Coast.

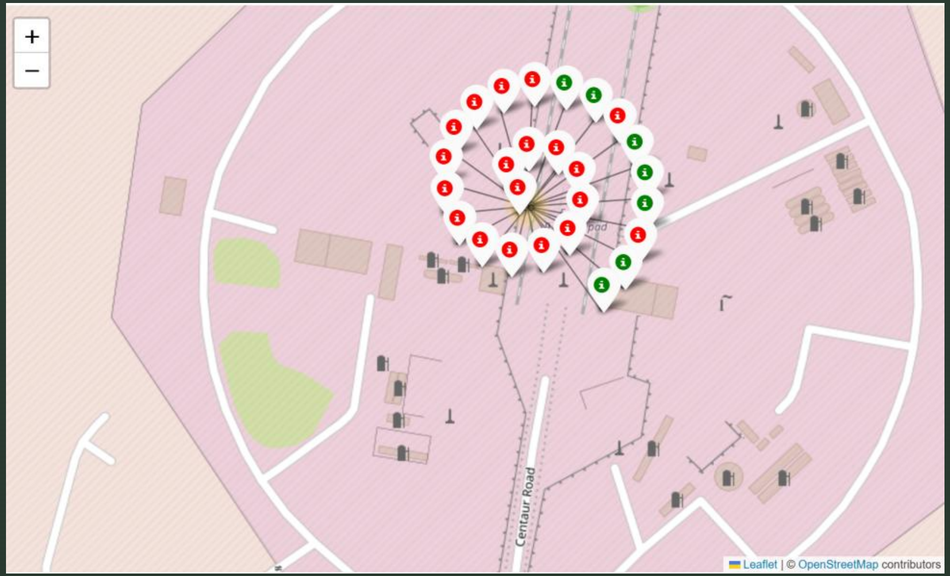


https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

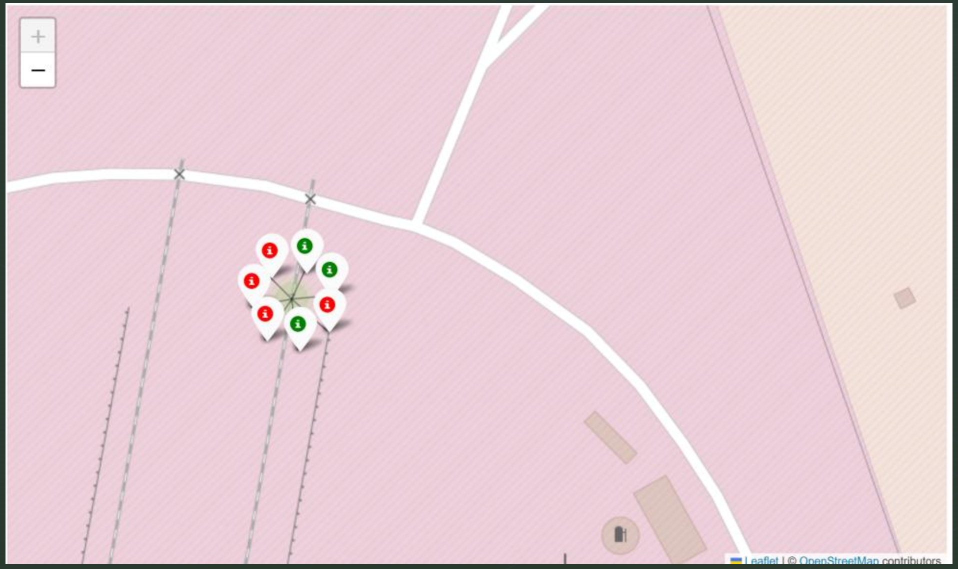
Folium Map Results

The following two maps show the success and failure of launches from CCAFS LC-40 and CCAFS SLC-40. Green markers indicate successful launches, while red markers indicate failures.
There were 26 launches from CCAFS LC-40 and 7 launches from CCAFS SLC-40, showing that the number of launches varied significantly between the two sites.

CCAFS LC-40



CCAFS SLC-40

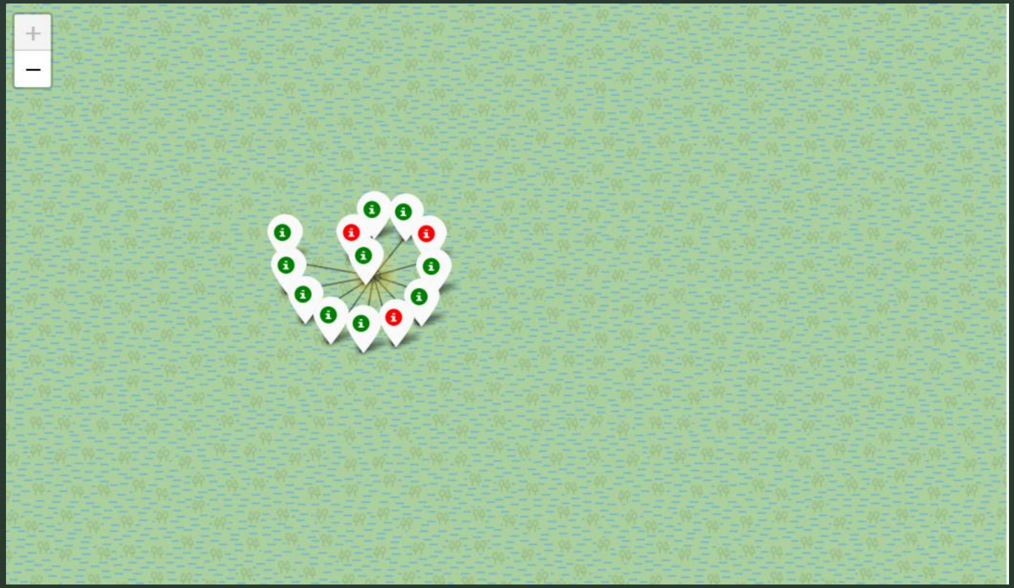


https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

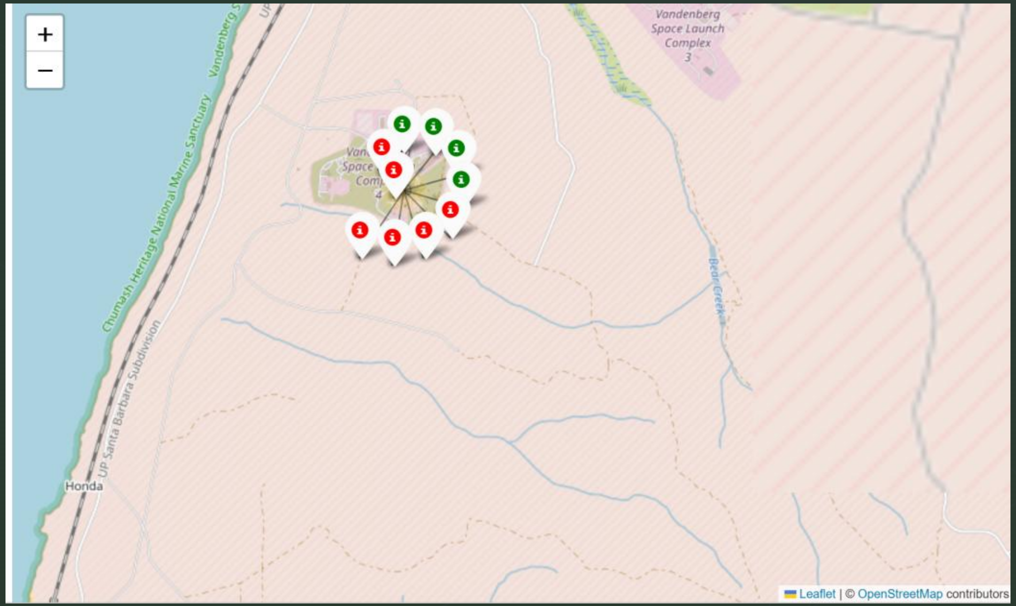
Folium Map Results

The following two maps show the success and failure of launches from KSC LC-39A and VAFB SLC-4E. Green markers indicate successful launches, while red markers indicate failures.
There were 13 launches from KSC LC-39A and 10 launches VAFB SLC-4E, showing that the number of launches varied significantly between the two sites.

KSC LC-39A



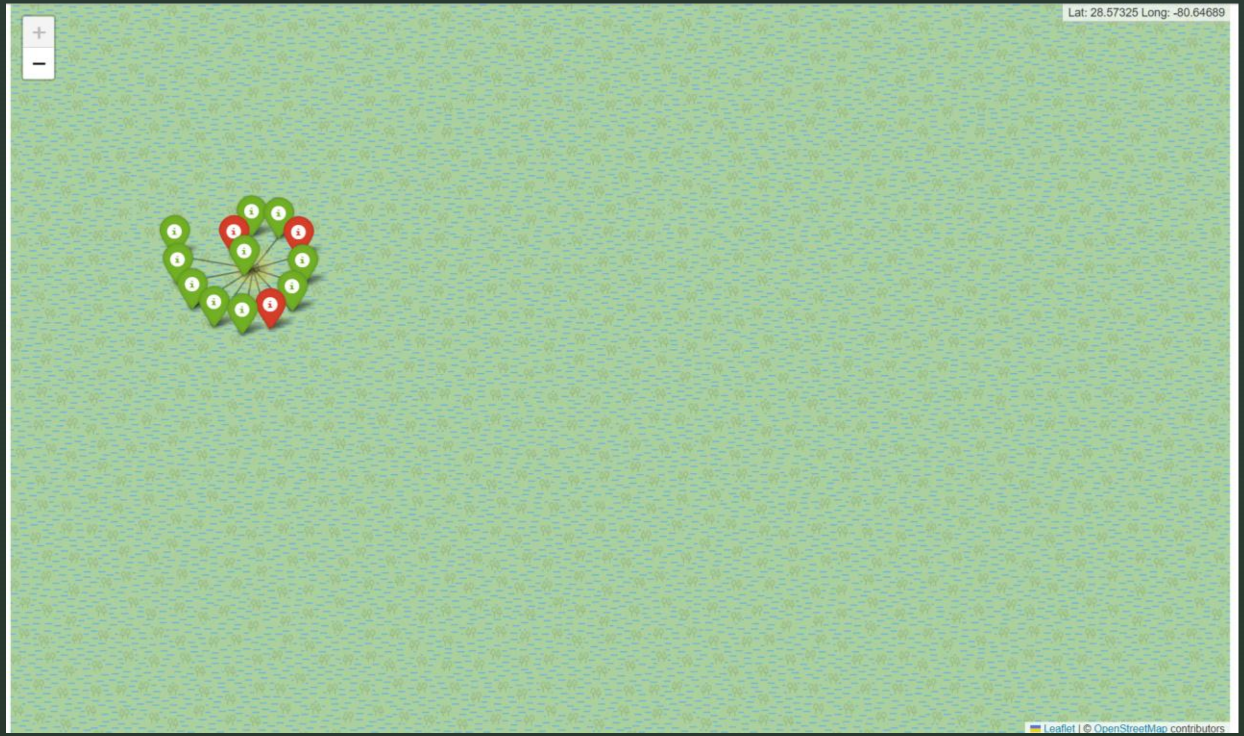
VAFB SLC-4E



https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

Folium Map Results

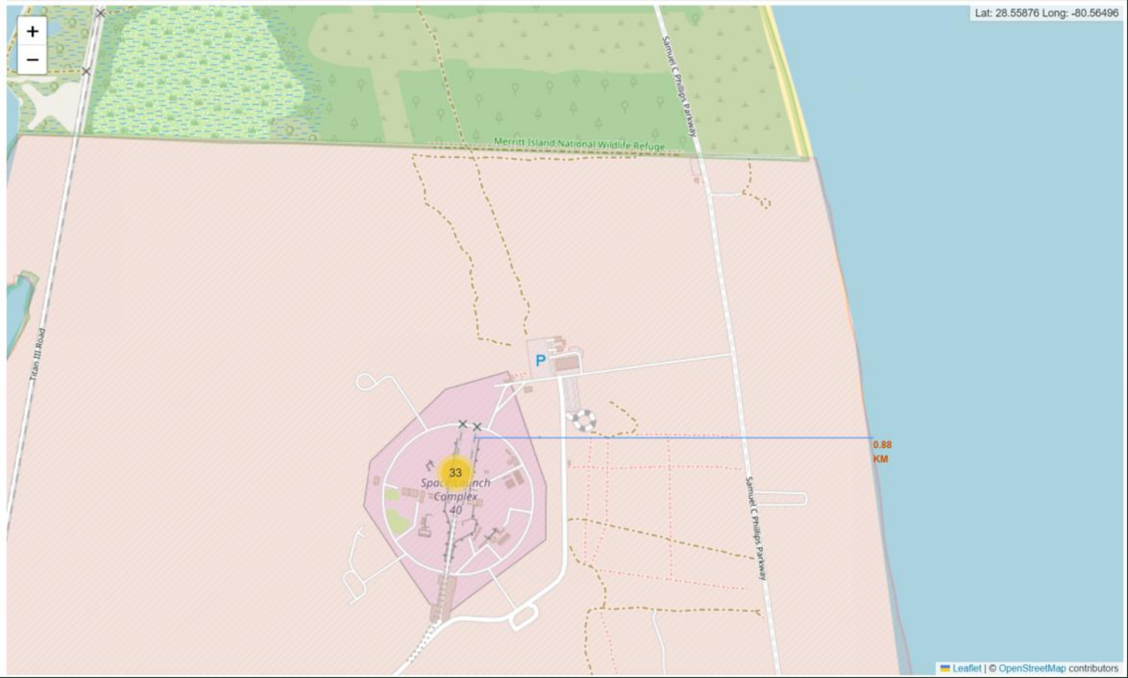
Folium was used to display the **latitude and longitude** of the mouse pointer position.
The coordinates are shown in the **top-right corner** of the map.
For example, **KSC LC-39A** is located at **28.57325° latitude and -80.64689° longitude**.
This makes it easier to identify the **exact coordinates** of each launch site, rather than viewing it only as a location on the map.



https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

Folium Map Results

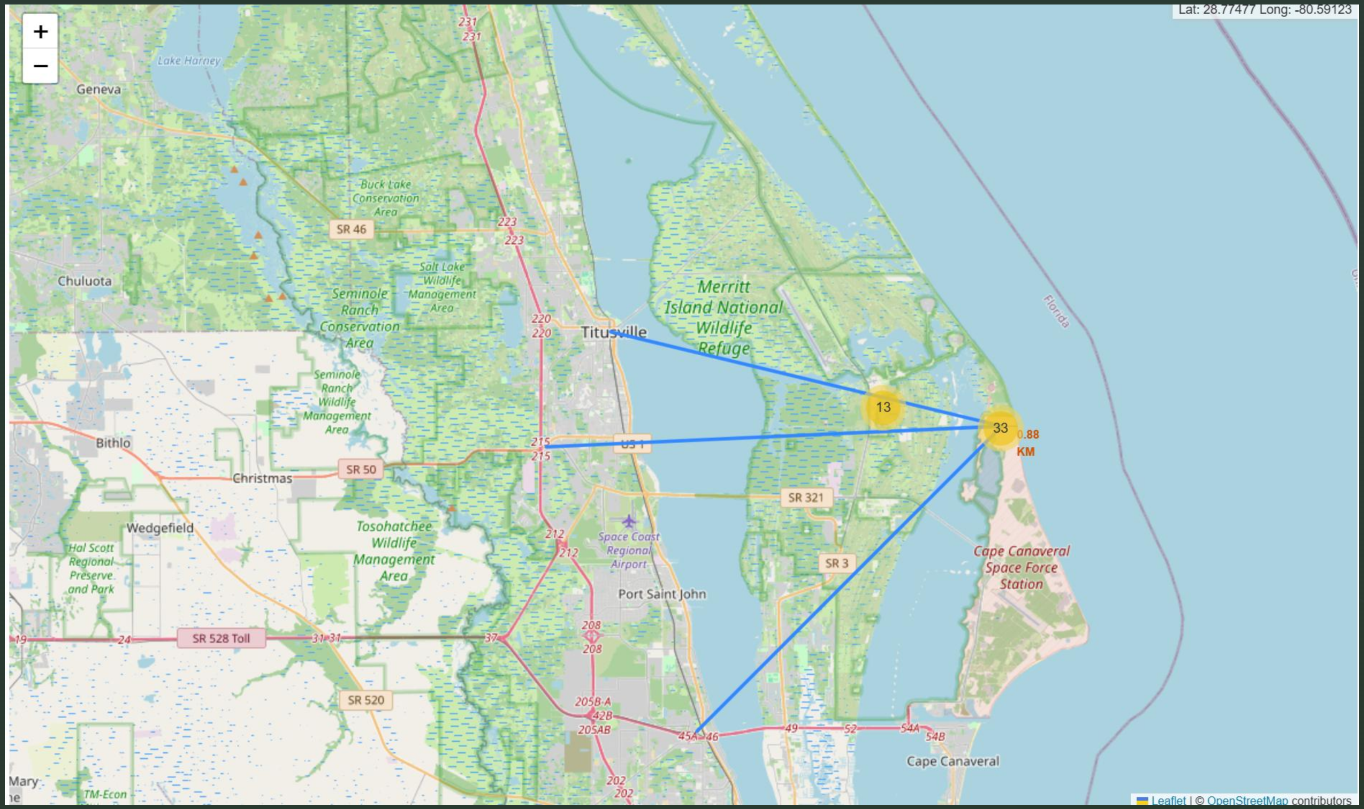
By using Folium, it is also possible to measure the distance between two locations.
This shows the distance from CCAFS SLC-40 to the nearest coast. We can see that the nearest coast is 0.88 km away from the launch site.
It allows us to see how far the launch site is from the coast.



https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

Folium Map Results

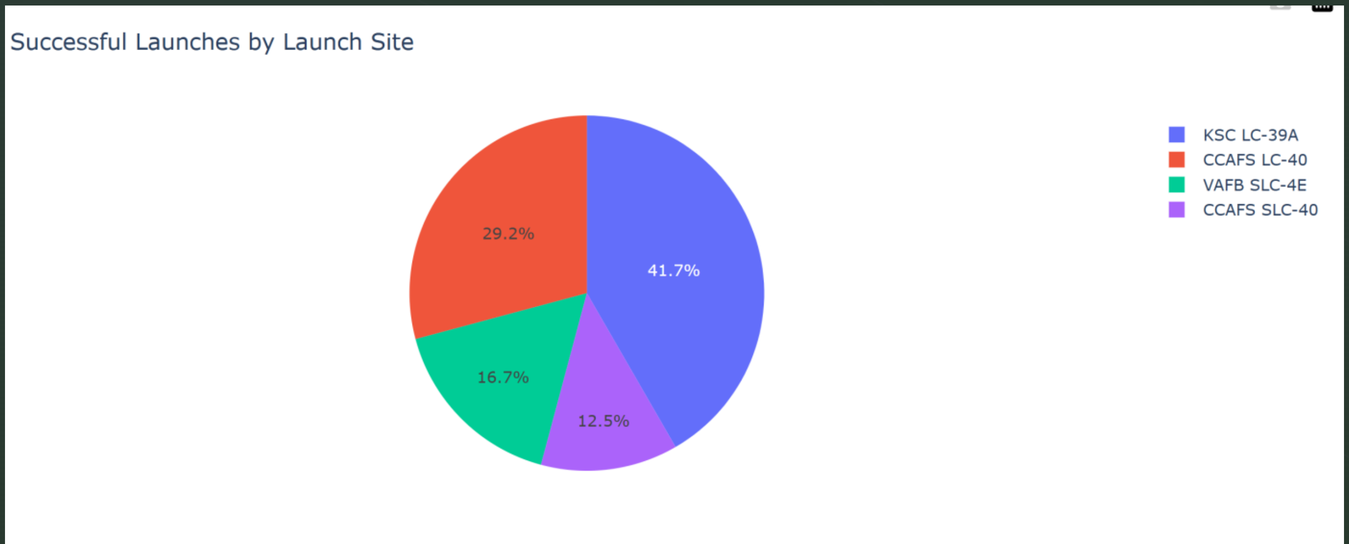
We drew lines from the launch site, CCAFS SLC-40, to the following three locations: Titusville, Florida East Coast Railway, and Cheney Highway. This allows us to visually confirm how far the launch site is from each location.



https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

Plotly Dash Dashboard（1）

I used Plotly Dash to create an interactive graph.
This pie chart shows the proportion of all successful launches that were carried out from each specific launch site.

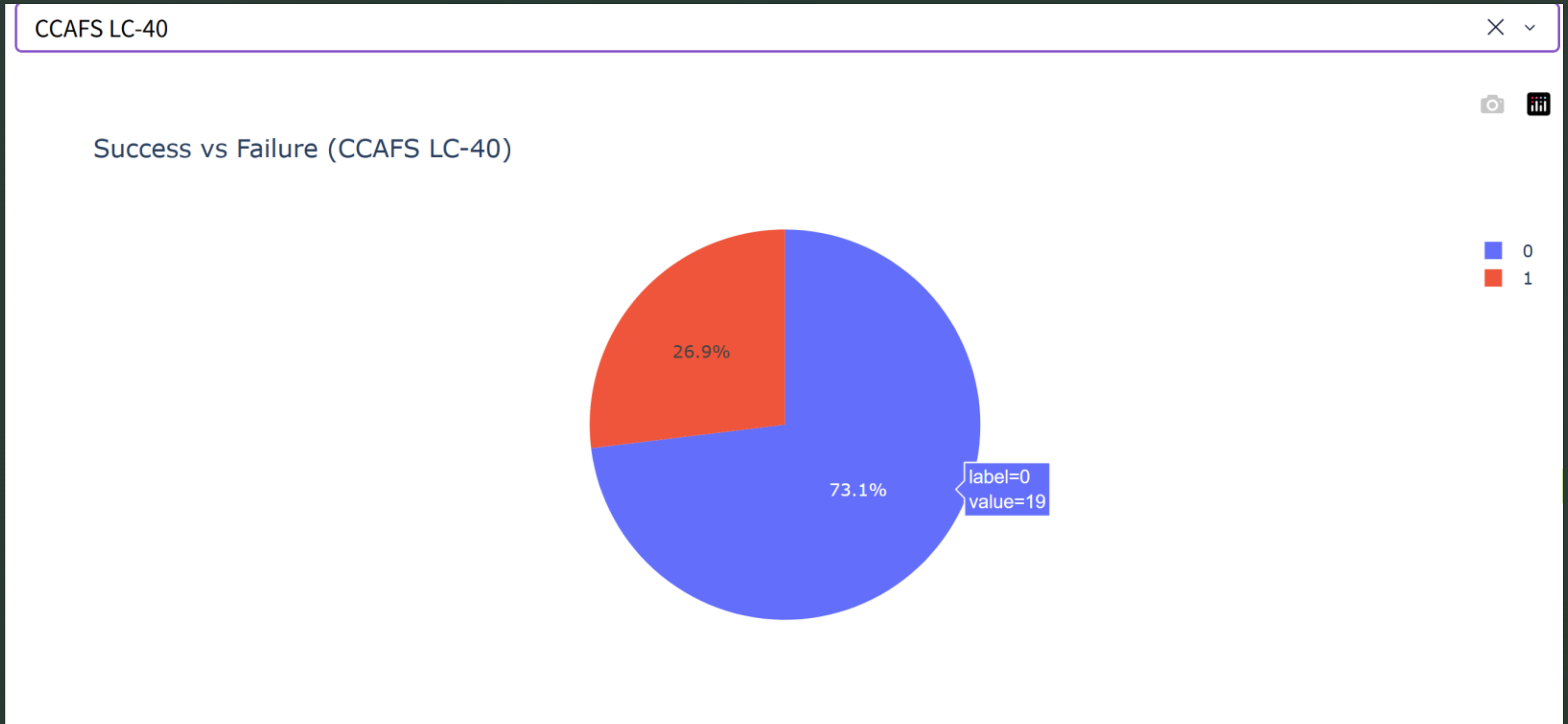


KSC LC-39A recorded the highest number of successful launches, accounting for 41.7% of all successful launches. This was followed by CCAFS LC-40 at 29.2%, VAFB SLC-4E at 16.7%, and CCAFS SLC-40 at 12.5%.

http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard（1)

This pie chart shows the success rate on CCAFS LC-40:

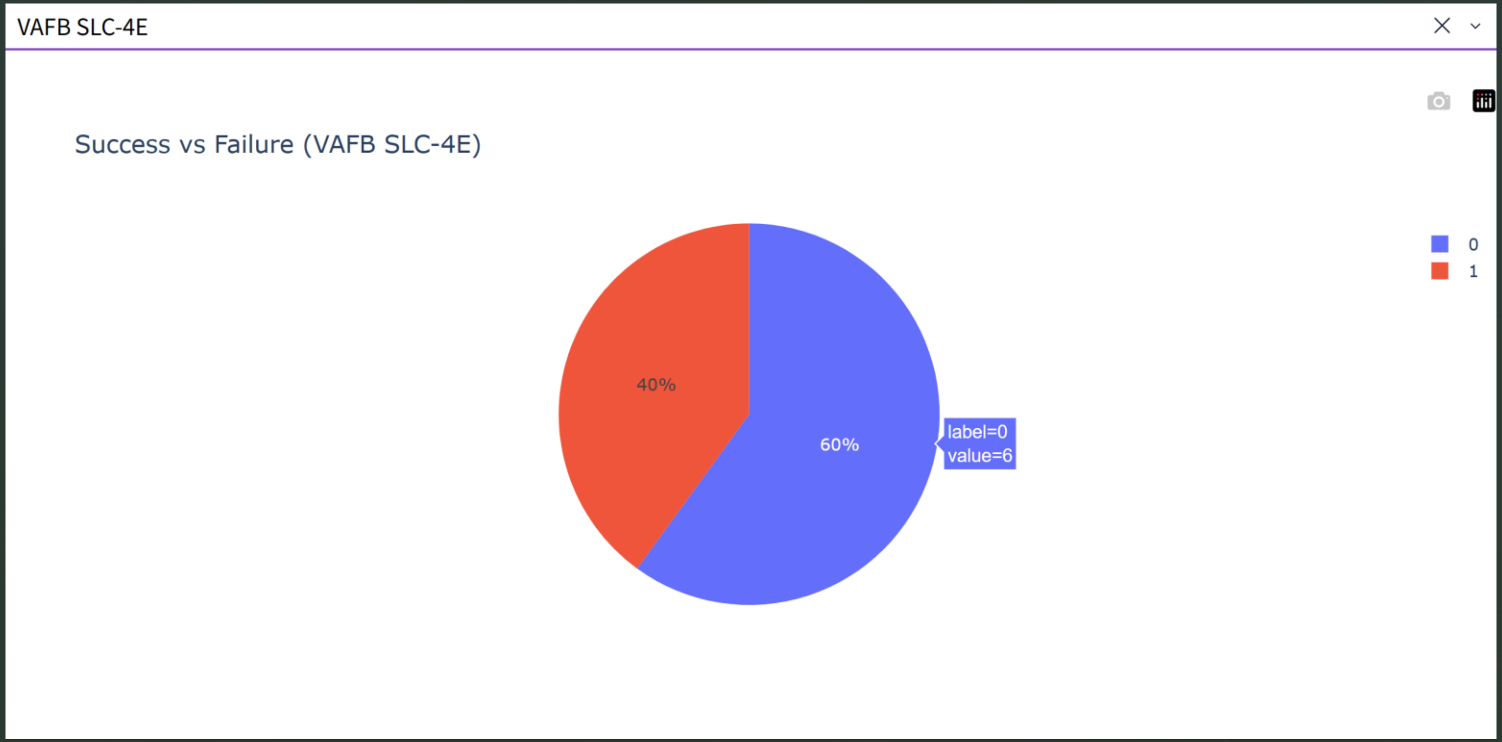


A value of 1 represents a successful launch.
19 successful launches were recorded, resulting in a success rate of 73.1%.

http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard（1）

This pie chart shows the success rate on VAFB SLC-4E:

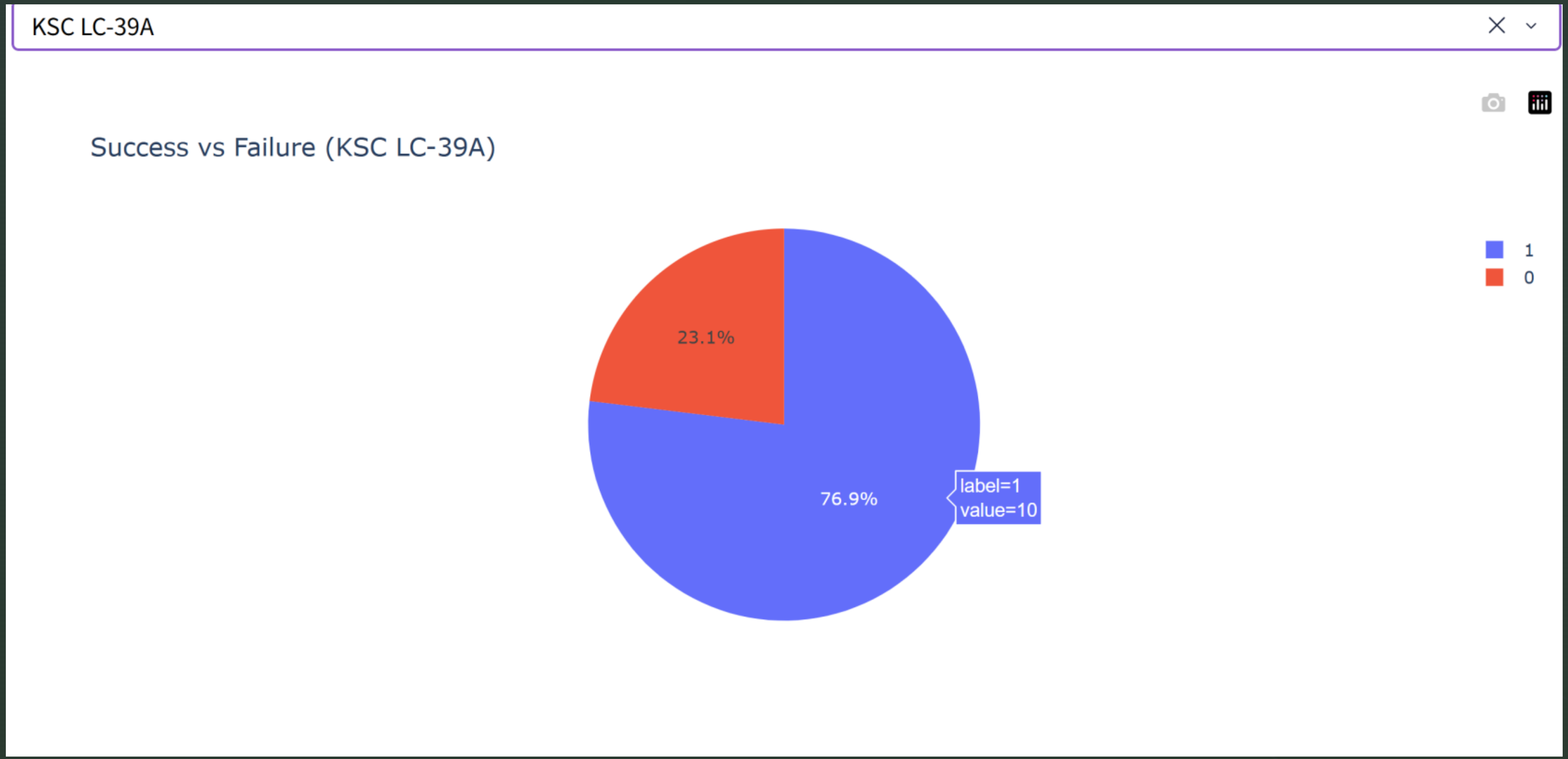


A value of 1 represents a successful launch.
Six successful launches were recorded, resulting in a success rate of 60%.

http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard（1)

This pie chart shows the success rate on KSC LC-39A:

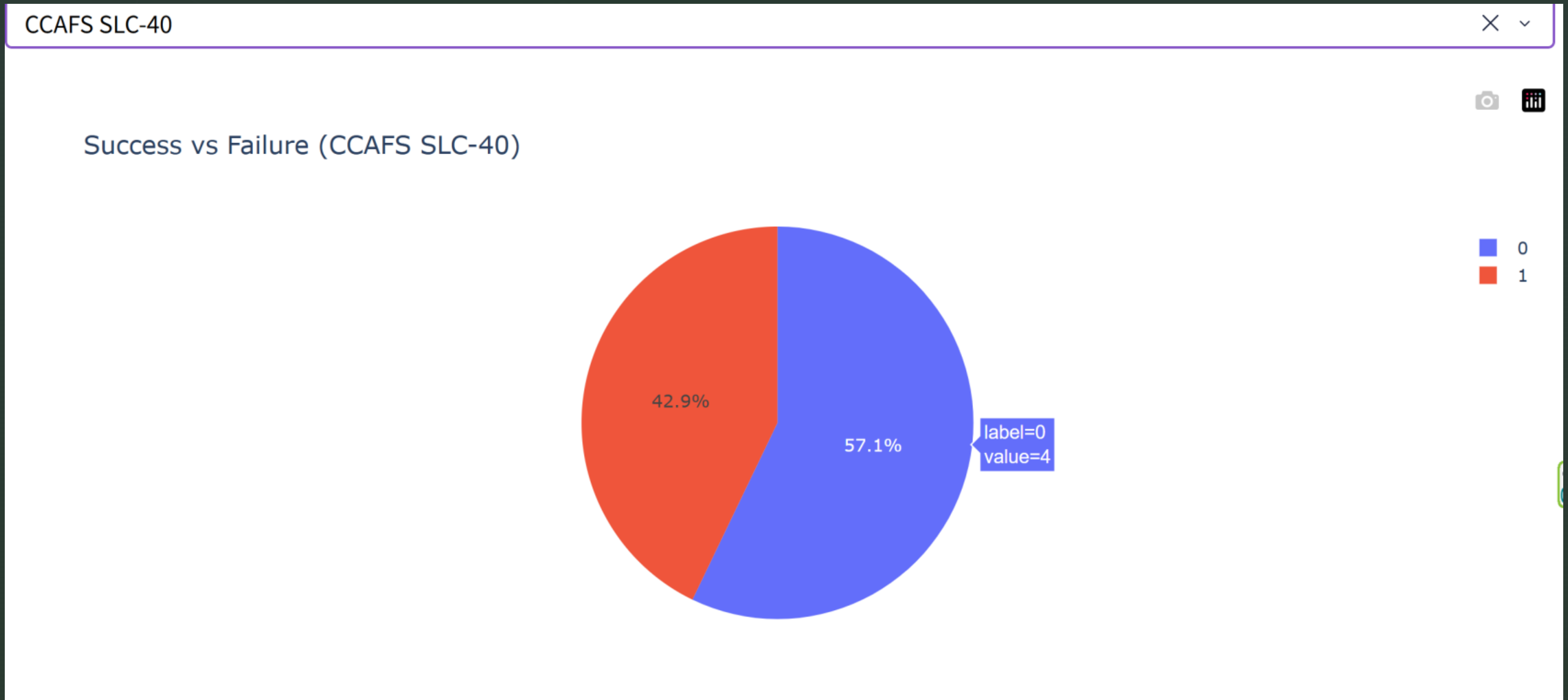


A value of 1 represents a successful launch.
Ten successful launches were recorded, resulting in a success rate of 76.9%.

http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard (1)

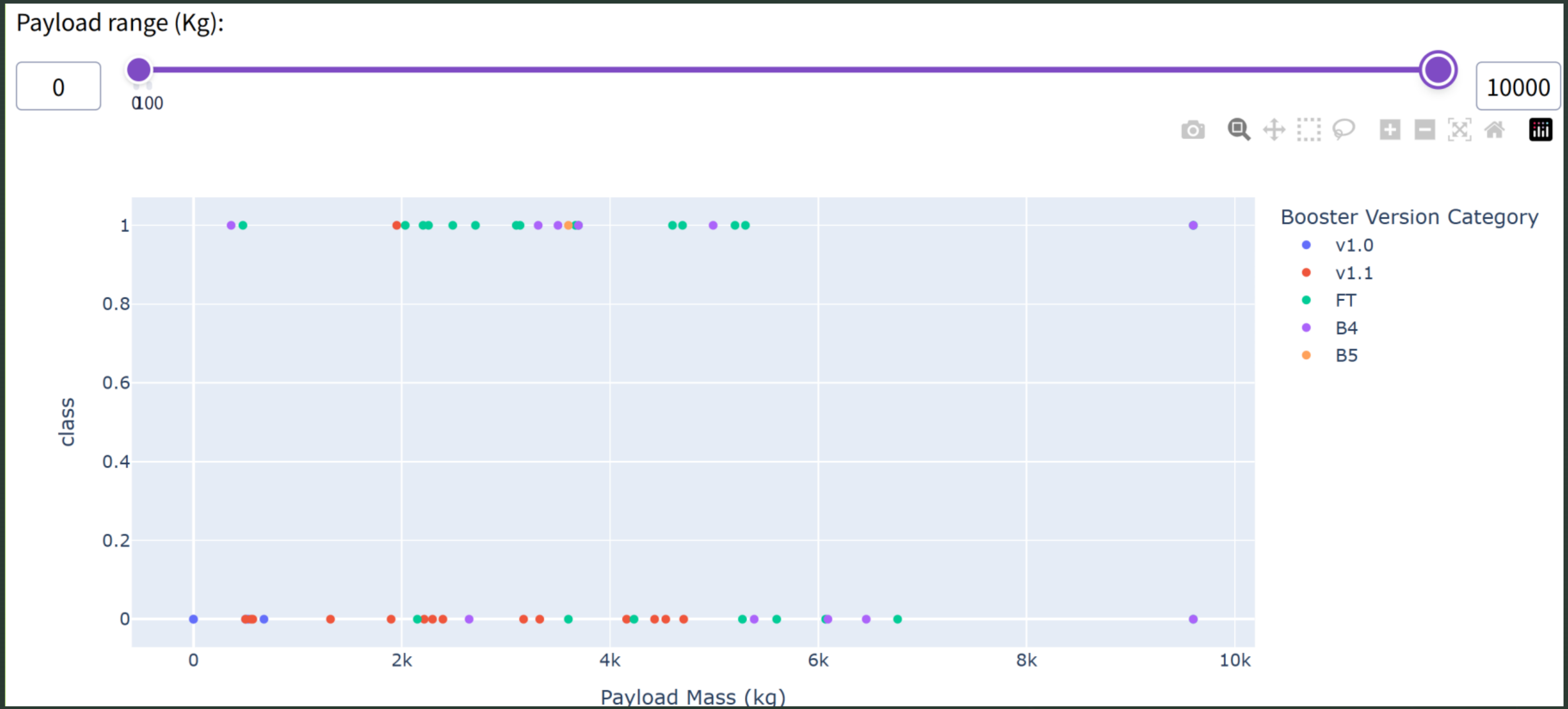
This pie chart shows the success rate on CCAFS SLC-40:



A value of 1 represents a successful launch.
Four successful launches were recorded, resulting in a success rate of 57.1%.

http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard (2)

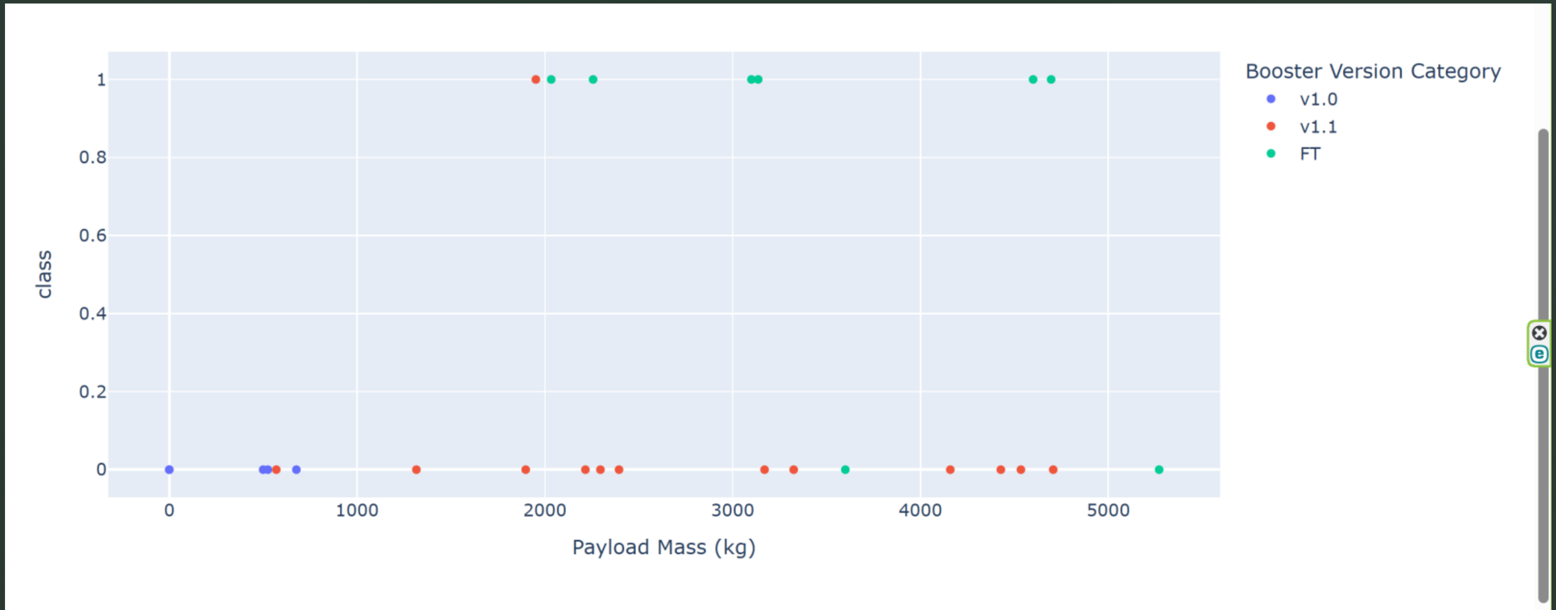


By booster version, FT and B4 have the highest number of successful launches, while failures are more common with V1.1. Launches carrying more than 8,000 kg used the B4 booster.

http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard (2)

This scatter plot shows the relationship between payload mass and launch success for each booster version at **CCAFS LC-40**. For payloads above 2,000 kg, the use of the FT version is more prominent. Meanwhile, the V1.0 version is frequently used for payloads below 1,000 kg, but no successful launches are recorded for this version. The V1.1 version is used across a wide range of payload masses. Overall, among the successful launches, the FT version appears most frequently.



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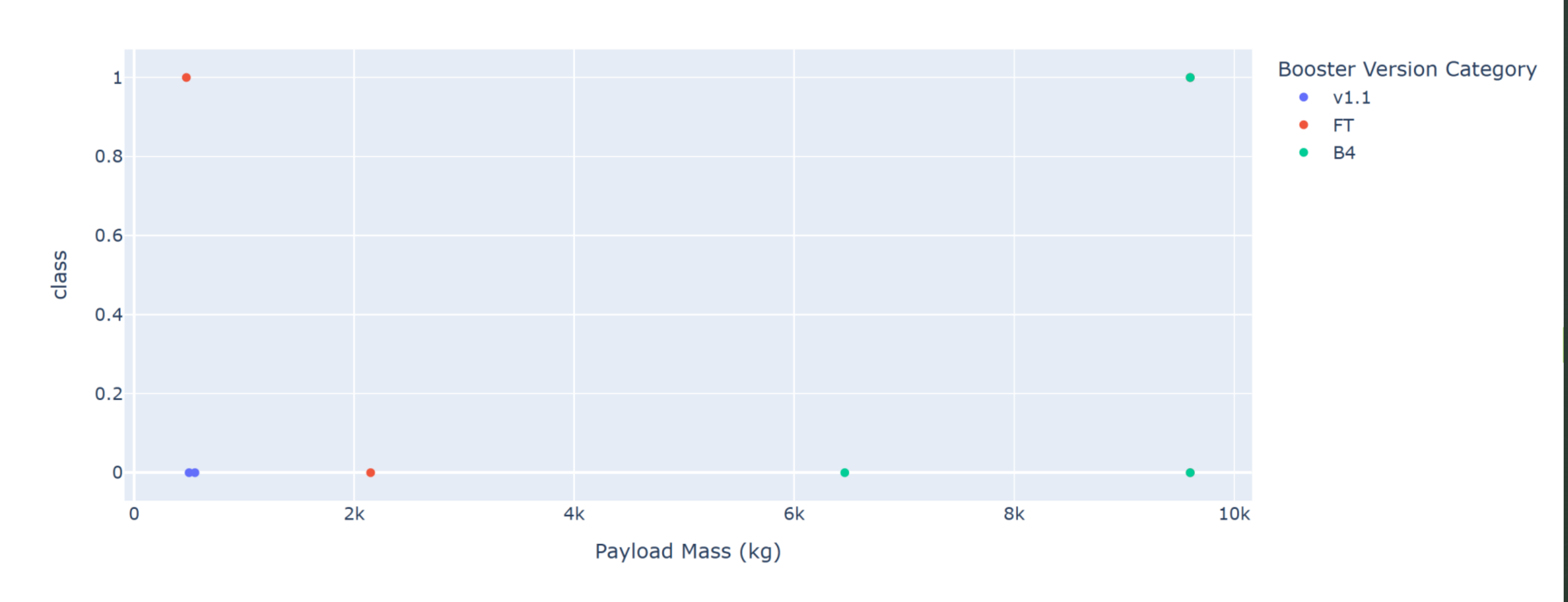
Plotly Dash Dashboard (2)

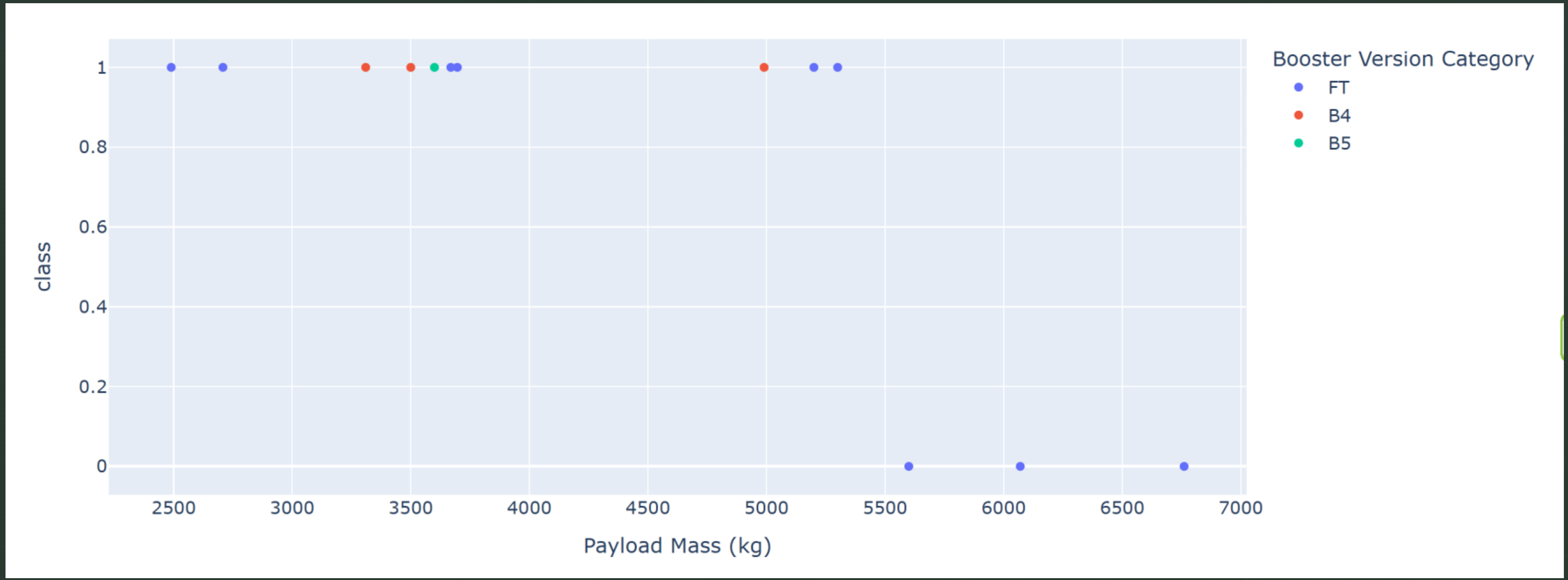
This scatter plot shows the relationship between payload mass and launch success for each booster version at **VAFB SLC-4E**. The V1.1 booster is observed in launches with payloads below 1,000 kg, and all of these attempts appear to have been unsuccessful. The FT booster was successful in attempts with payloads below 1,000 kg, but unsuccessful in attempts with payloads above 2,000 kg. The B4 booster was mainly used for payloads above 6,000 kg, with the highest payload ranging from 9,000 kg to less than 10,000 kg, and one of these attempts was successful.



http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard (2)

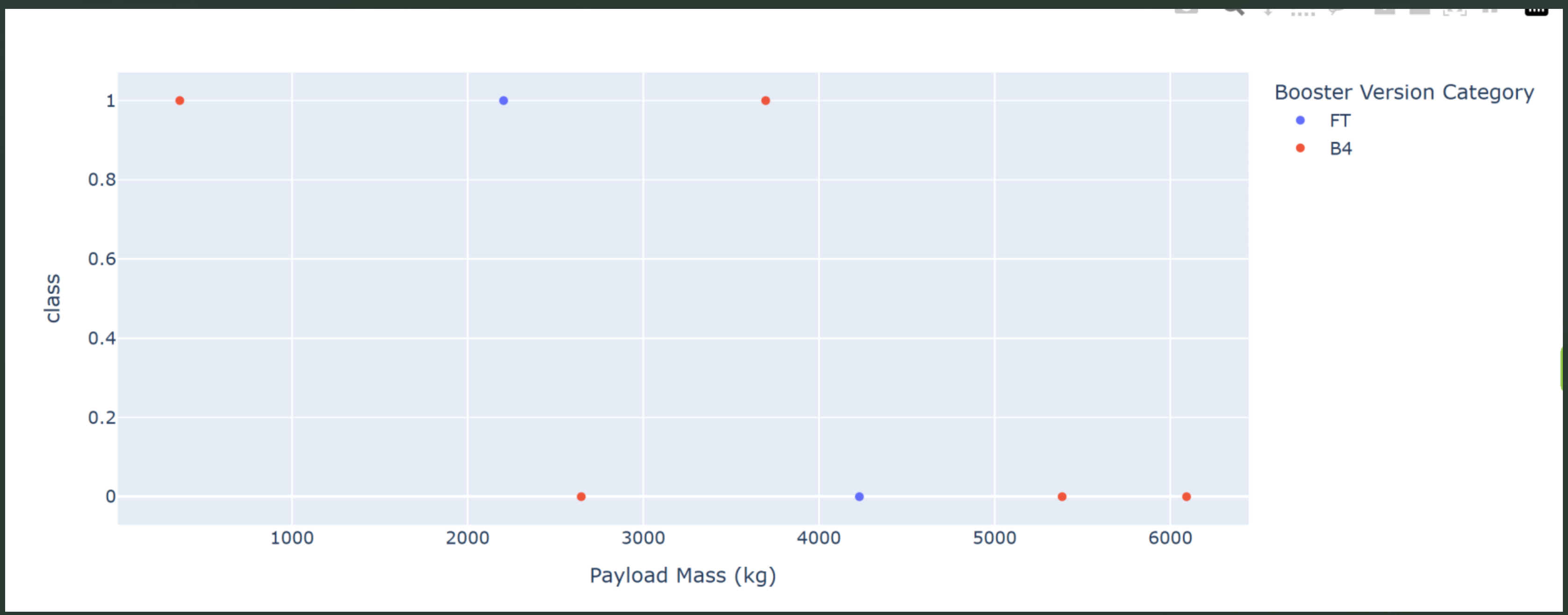
This scatter plot shows the relationship between payload mass and launch success for each booster version at **KSC LC-39A**. At this site, a trend can be observed in which launches with payloads below 5,500 kg tend to be successful, while those above 5,500 kg tend to be unsuccessful. The use of B5 is limited to launches with payloads of around 3,500 kg, while B4 is observed in launches with payloads between 3,000 and 5,000 kg. FT is observed multiple times across a wider range, from 2,500 to 7,000 kg; however, it appears that launches with payloads exceeding 5,500 kg were unsuccessful.



http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Plotly Dash Dashboard (2)

This scatter plot shows the relationship between payload mass and launch success for each booster version at **CCAFS SLC-40**. At this site, two booster versions, FT and B4, were used. FT was used for payloads ranging from 2,000 kg to 4,000 kg, while B4 was used across a wider range, from below 1,000 kg to over 6,000 kg. It can be observed that launches with lighter payloads tended to have a higher number of successful launches.



http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Predictive Analysis Methodology

1. Data Preprocessing

Class was used as the target variable.
Features were standardized using StandardScaler.
The data was split into training and test sets (80% / 20%).

2. Model Training

Logistic Regression
Support Vector Machine (SVM)
Decision Tree
K-Nearest Neighbors (KNN)

3. Hyperparameter Tuning

GridSearchCV with 10-fold cross-validation was used to select the optimal hyperparamerset.

4. Model Evaluation

Test accuracy

Confusion matrix

5. Model Comparison

The performance of the four classification models was compared using test-set accuracy.

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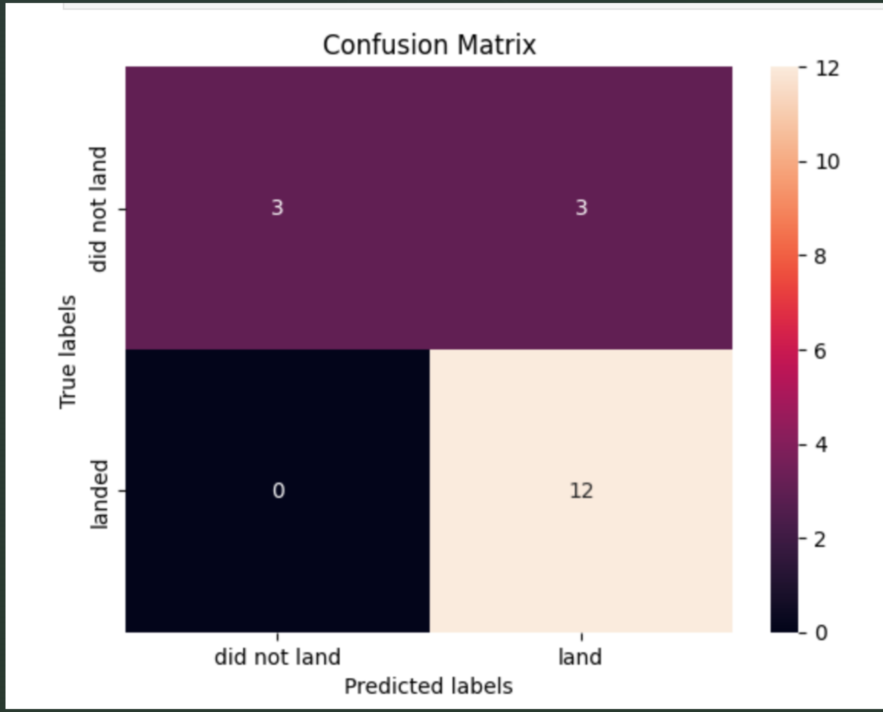
Predictive Analysis Results



<u>Logistic Regression — Results</u>

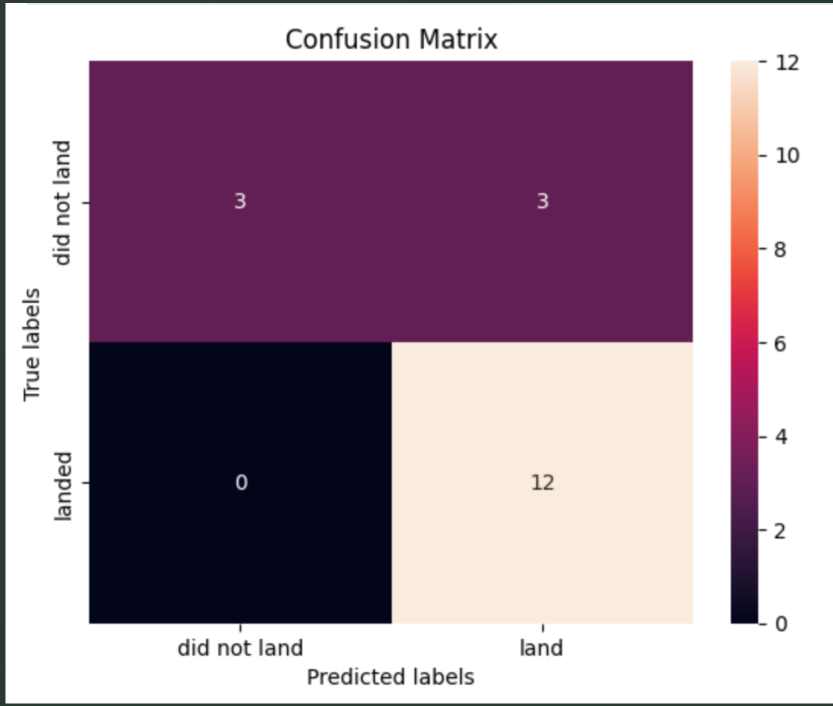
Best parameters: C = 0.01, penalty = L2, solver = lbfgs
Validation accuracy: 84.6%
Test accuracy: 83.3%

For the **18 test predictions**:

- **15 launches were predicted to land**. Of these, **12 actually landed**, while **3 did not**.
- **3 launches were predicted not to land**, and **all 3 actually did not land**.

https://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Predictive Analysis Results



SVM— Results

Best parameters: C = 1.0, gamma = 0.03162277660168379, kernel = sigmoid
Validation accuracy: 84.8%
Test accuracy: 83.3%

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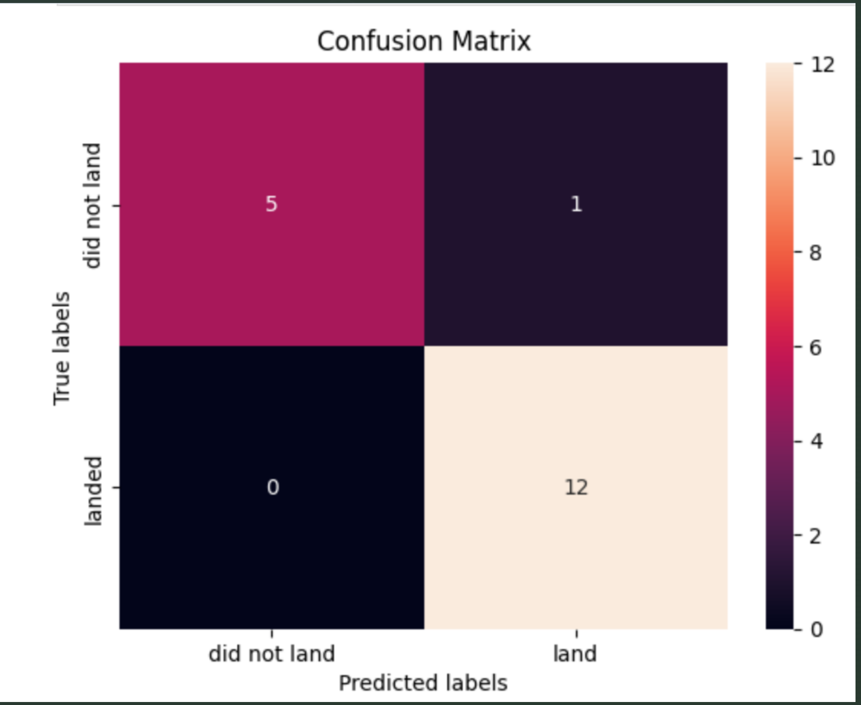
https://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Predictive Analysis Results



<u>Tree Decision— Results</u>

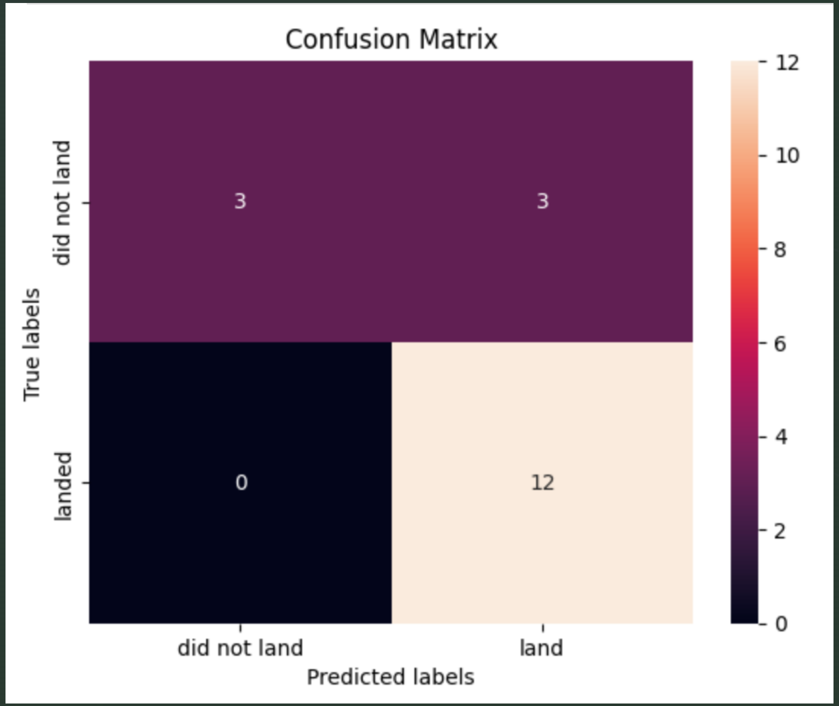
Best parameters: criterion = gini, max_depth = 4, max_features = sqrt, min_samples_leaf = 2, min_samples_split = 10, splitter = random
Validation accuracy: 87.5%
Test accuracy: 94.4%

For the **18 test predictions**:

13 launches were predicted to land. Of these, **12 actually landed**, while **1 did not**.
5 launches were predicted not to land, and **all 5 actually did not land**.

https://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Predictive Analysis Results



<u>KNN— Results</u>

Best parameters: algorithm = auto, n_neighbors = 10, p = 1
Validation accuracy: 84.8%
Test accuracy: 83.3%

For the **18 test predictions**:

- **15 launches were predicted to land**. Of these, **12 actually landed**, while **3 did not**.
- **3 launches were predicted not to land**, and **all 3 actually did not land**.

https://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Predictive Analysis Results

Both the **validation accuracy** and **test accuracy** were highest for the **Decision Tree model**, at **87.5%** and **94.4%**, respectively.
This suggests that the Decision Tree model performed better than the other models for predicting landing success.

The following table shows the test accuracies:

Model	Accuracy
Logistic Regression	83.3%
SVM	83.3%
Decision Tree	94.4%
KNN	83.3%

Overall, the Decision Tree performed best on the test data.

https://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Conclusion

The analysis indicates that Falcon 9 launch success is influenced by several key factors, including launch site, orbit, payload mass, and booster version.

Exploratory and geographical analyses revealed significant differences in launch success rates across various sites and booster versions, highlighting the importance of these characteristics in operational strategies.

Furthermore, machine learning analysis demonstrated that the Decision Tree model outperformed other models, achieving an impressive accuracy of **94.4%** on the test data.

Overall, this analysis suggests that both launch characteristics and booster technology play critical roles in predicting Falcon 9 landing success. However, it is important to note that the dataset is relatively small, with few records, so the results should be interpreted with caution. Future work should aim to incorporate a larger dataset and explore additional variables that might further enhance predictive capabilities.

Innovative Insights / Recommendations

Innovative Highlights:

This analysis utilized a combination of machine learning models, including Logistic Regression, SVM, Decision Tree, and KNN, to predict Falcon 9 landing success. The Decision Tree model's high accuracy of **94.4%** represents a significant advancement in predictive analysis for space missions.

Geographical analysis using Folium revealed spatial patterns in launch success, highlighting the importance of site selection based on payload and orbit type.

Recommendations:

Data Expansion: To enhance the robustness of the analysis, future studies should aim to incorporate a larger dataset that includes more launch attempts and diverse payload types. This would provide a more comprehensive understanding of factors influencing success rates.

Exploration of Additional Variables: Future analyses could benefit from exploring additional variables, such as weather conditions or recent technological advancements in booster design, which may further impact launch success.

Continuous Monitoring: Implementing a system for continuous monitoring and updating of launch data would allow for real-time analysis and more accurate predictions, helping to refine SpaceX’s operational strategies.

Access to my GitHub

https://github.com/HaruNNH/Final-Projects